

5 Recombinant DNA technology is used to make recombinant human proteins.

Two of the available methods to obtain the gene of interest are:

- cutting the gene out of genomic DNA using restriction enzymes
- obtaining messenger RNA (mRNA) from cells that are expressing the gene and then using reverse transcriptase to make complementary DNA (cDNA).

Plasmids can be used as vectors to transfer the gene of interest into a host organism.

(a) Recombinant human insulin is a protein that is made using recombinant DNA technology.

Bacteria can be used as host cells to express the recombinant protein.

For the human insulin gene to be successfully expressed in bacteria, one method chosen to obtain the gene is to extract mRNA from β -cells in the pancreas.

The gene coding for insulin is not expressed in the bacterial host when it has been obtained by cutting it out of genomic DNA.

Suggest **and** explain how the structural difference of cDNA and genomic DNA leads to only cDNA being expressed successfully.

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- (b) Plasmids are cut using a restriction enzyme to create sticky ends. The plasmids are mixed with many copies of the desired gene and DNA ligase. The gene is inserted into many plasmids.

(i) State the role of DNA ligase in the formation of recombinant plasmids.

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(ii) Explain why a promoter, as well as the gene, may have to be transferred into the plasmid.

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(c) Explain how a marker gene coding for a fluorescent product could be used.

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(d) The unicellular fungus *Saccharomyces cerevisiae* is a species of yeast that has been used to produce human insulin. *S. cerevisiae* cells are able to take up recombinant plasmids.

Suggest advantages of using yeast compared to using bacteria for human insulin production.

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